

LEARNING TO EVALUATE STRUCTURED ARTIFACTS FROM HUMAN PREFERENCES

Anonymous authors

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ABSTRACT

Evaluating generated artifacts requires more than assigning a score: an evaluator must obtain relevant evidence, apply task-specific criteria, and reconcile competing aspects of quality. A disagreement with human preference may arise at any of these stages, making the evaluator itself a useful object of optimization. We study learning executable evaluation programs from human overall preferences while keeping the underlying model fixed. We introduce ERA, Evidence-Guided Revision of Agentic Evaluators, as the method for revising these programs. An evaluation program comprises artifact representations, rubrics, specialist evaluators, evidence tools, routing, and score synthesis. ERA uses artifact-level execution evidence to formulate revision hypotheses, constructs bounded program edits, and assesses candidates on a separate training cohort. At deployment, the resulting evaluator scores each artifact independently, without access to its opponent or preference labels. We specify an evaluation protocol that separates adaptive development from terminal validation and held-out testing, and tests both preference agreement and the usefulness of evaluator feedback for artifact refinement. The proposed experiments assess preference agreement, fixed-pool selection, and feedback-guided refinement under matched models and budgets.

1 INTRODUCTION

An automatically generated slide deck can preserve the source facts yet obscure their relationships through poor visual hierarchy. A reconstructed webpage can resemble a reference image while failing to implement a required interaction. These examples illustrate a difficulty in evaluating structured artifacts: different quality judgments require different evidence, and an overall preference depends on how those judgments are combined. Recent benchmarks for presentations and front-end generation make such evaluation requirements explicit (Zheng et al., 2025; Chen et al., 2026; Si et al., 2025).

Language-model evaluators provide a flexible alternative to fixed similarity metrics, but their agreement with people depends on the instructions, evidence, and decision procedure they receive (Liu et al., 2023; Zheng et al., 2023; Shankar et al., 2024). Human feedback can identify a preferred artifact without identifying why an evaluator disagreed. A missing observation, an inappropriate criterion, and an incorrect trade-off can produce the same ranking error. Merely increasing the weight of a criterion does not resolve all three causes.

We study the following question: *can human overall preferences guide the revision of an executable evaluator, including how it acquires and combines evidence?* We formulate the evaluator as a structured program and introduce ERA as the method for revising it. The program exposes six connected components: artifact representation, rubric, specialist evaluators, evidence tools, evidence routing, and synthesis. An outer revision process uses authorized training examples and execution traces to identify a candidate mechanism of error and propose a corresponding program change. The deployed program remains a pointwise evaluator: it receives one artifact and its task context, not the human comparison used to develop it.

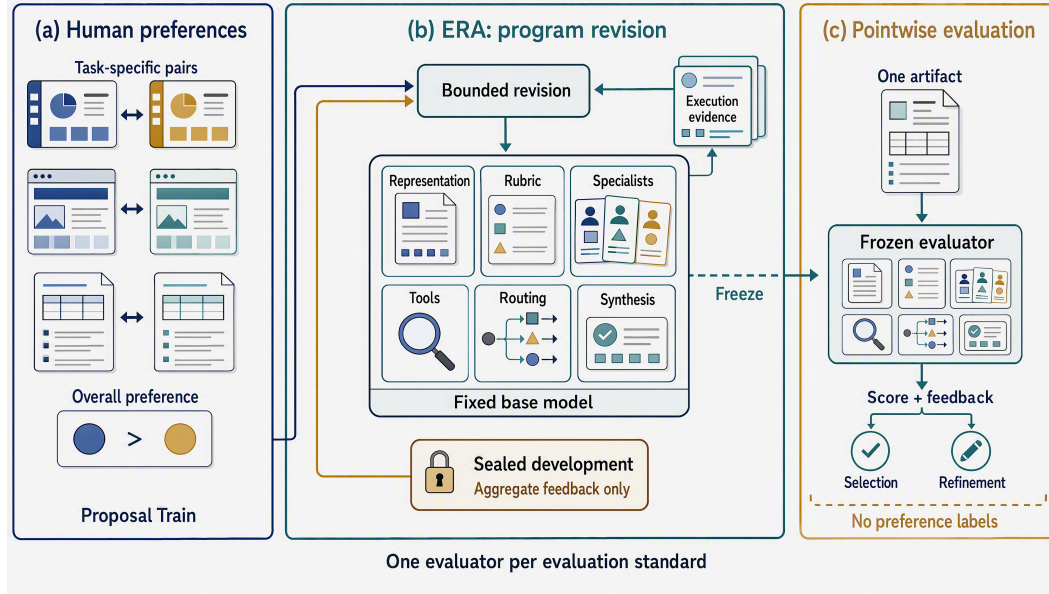


Figure 1: Learning evaluation programs from human preferences. Panel (a) supplies training comparisons within an evaluation standard. Panel (b) uses ERA to revise the program from execution evidence and aggregate development feedback. Panel (c) uses the frozen evaluator for independent scoring and feedback; selection and refinement are evaluation targets, not measured outcomes here. The generated artifact miniatures illustrate possible applications, not benchmark cases. Each evaluation standard has its own learned evaluator.

Figure 1 separates training supervision, program revision, and use of the frozen evaluator. The same revision method can be instantiated under different evaluation standards, without assuming that one learned scorer serves all domains.

Reflective optimization and automated agent design support revisions to prompts, workflows, and code (Agrawal et al., 2026; Hu et al., 2025; Zhang et al., 2025b;a). Our focus is the evaluator-specific learning problem: overall preference provides supervision, execution evidence supports revision, and independent artifact scores must preserve useful quality distinctions. The central hypothesis is that revising evidence acquisition and synthesis can address errors that remain under prompt-only adaptation. This is an empirical hypothesis, not an assumed benefit of a larger search space.

The paper develops three elements: (i) a structured formulation of the evaluator-revision problem; (ii) an evidence-guided search procedure with separate proposal, measurement, and allocation decisions; and (iii) a controlled protocol for assessing preference agreement, artifact selection, and feedback-guided refinement.

2 RELATED WORK

Learning and assessing evaluators. G-Eval and MT-Bench study model-based judgments and their relationship to human evaluation (Liu et al., 2023; Zheng et al., 2023). Prometheus and Prometheus 2 train models for rubric-conditioned evaluation, while Self-Taught Evaluators use synthetic supervision to improve judges (Kim et al., 2024a;b; Wang et al., 2024). EvalGen emphasizes that evaluation criteria themselves require alignment with human preferences (Shankar et al., 2024). RewardBench and JudgeBench test evaluators on challenging comparisons, including cases where preference and objective correctness require careful distinction (Lambert et al., 2024; Tan et al., 2025). ERA instead holds model weights fixed and revises the executable evaluation procedure. Overall human preference is its supervision target; inferred explanations are not additional human annotations.

Optimizing language-model programs. DSPy exposes compositional language-model pipelines to optimization (Khattab et al., 2024). ProTeGi, OPRO, TextGrad, and GEPA use model-generated feedback or proposals to improve instructions and programs (Pryzant et al., 2023; Yang et al., 2024; Yuksekgonul et al., 2025; Agrawal et al., 2026). ADAS, AFlow, and the Darwin Gödel Machine further investigate the automated construction or revision of agentic systems (Hu et al., 2025; Zhang et al., 2025b;a). Their broad search spaces make them relevant baselines for assessing an evaluator-specific representation and evidence interface. ERA specializes these interfaces to evaluation, where a ranking failure must be related to observations, criteria, and synthesis.

Artifact quality and downstream use. Evaluation standards differ across domains. SummEval and BERTScore address summarization evaluation and text similarity (Fabbri et al., 2021; Zhang et al., 2020); ImageReward, Pick-a-Pic, and GenEval capture different aspects of image preference and text-image alignment (Xu et al., 2023; Kirstain et al., 2023; Ghosh et al., 2023). Such signals may supply evidence or comparisons, but are not interchangeable definitions of overall quality. ReAct, Reflexion, and Self-Refine show ways to combine observations, reflection, and revision (Yao et al., 2023; Shinn et al., 2023; Madaan et al., 2023). Preference-based summarization, instruction tuning, and direct preference optimization connect evaluation to model learning (Stiennon et al., 2020; Ouyang et al., 2022; Rafailov et al., 2023). The proposed evaluation focuses on an evolved evaluator as a selector and critic; reward-guided parameter training is outside the empirical scope.

3 PROBLEM FORMULATION

The learning problem is defined within a domain, meaning a shared evaluation standard rather than an artifact format. The revision method can be instantiated with different domain adapters and criteria; each learned evaluator remains specific to its evaluation standard.

Let x denote a task with its permitted context and references, and y a generated artifact. Training comparisons (x_i, y_i^+, y_i^-) indicate an overall human preference for y_i^+ over y_i^- . This supervision, denoted H_0 , need not include dimension-level ratings or explanations. The superscripts identify the human ordering in the learning objective only; they are never supplied to the deployed scorer.

We represent an evaluator by

$$E_\theta = (A, K, V, T, \Pi, G), \quad (s_\theta, \tau_\theta) = E_\theta(x, y), \quad (1)$$

where A constructs artifact views, K specifies evaluation criteria, V is a set of specialist evaluators, T provides evidence tools, Π routes evidence, and G synthesizes an overall score. The execution trace τ_θ records the observations and judgments that support the score. Model identity, runtime interfaces, and experimental constraints are fixed outside θ . The objective is to improve preference agreement on unseen tasks within that standard while retaining complete execution and an acceptable evaluation cost.

Pointwise, repeated measurement. Because model-based scoring is stochastic, each artifact receives $R \geq 3$ complete evaluations under the same frozen protocol. We first aggregate scores *within each artifact*, then compare:

$$\bar{s}_\theta(x, y) = \text{median}_{r=1, \dots, R} s_\theta^{(r)}(x, y), \quad z_i = \mathbb{I}[\bar{s}_\theta(x_i, y_i^+) > \bar{s}_\theta(x_i, y_i^-)]. \quad (2)$$

A predicted tie is incorrect for a strict human preference. This procedure is not majority voting over pairwise decisions. An incomplete execution does not yield a valid repeated-score estimate; coverage and missing-outcome bounds accompany accuracy (Appendix A). Human ties and ambiguous annotations require a separate, prespecified treatment.

Data access. We partition examples by their original task or source before constructing pairs. The proposal set $\mathcal{D}_{\text{prop}}$ exposes training artifacts, H_0 , and execution traces. A disjoint training development set $\mathcal{D}_{\text{dev}}$ supports repeated candidate measurement, returning only aggregate feedback to the optimizer. Validation $\mathcal{D}_{\text{val}}$ is used only after the candidate shortlist is frozen, and Test/OOD data remain locked until the method and analysis are fixed. Adaptively reused development accuracy is a search signal, not an unbiased estimate of generalization.

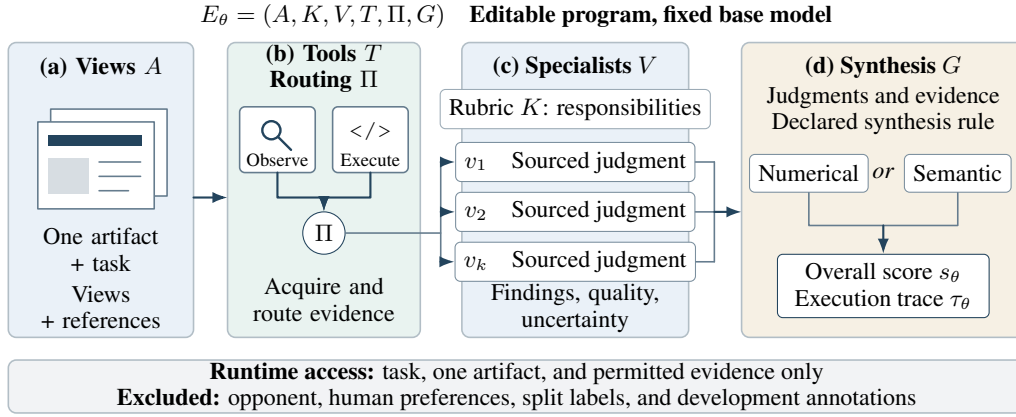


Figure 2: The executable evaluator revised by ERA. Panel (a) constructs task-relevant artifact views; (b) acquires and routes evidence; (c) applies the rubric through specialist judgments; and (d) synthesizes the overall score and trace using the declared numerical or semantic rule. All six components are editable, while the base model remains fixed. The artifact miniature is illustrative; preference labels and the comparison opponent never enter execution.

4 EVIDENCE-GUIDED EVALUATOR REVISION

4.1 AN EXECUTABLE EVALUATOR

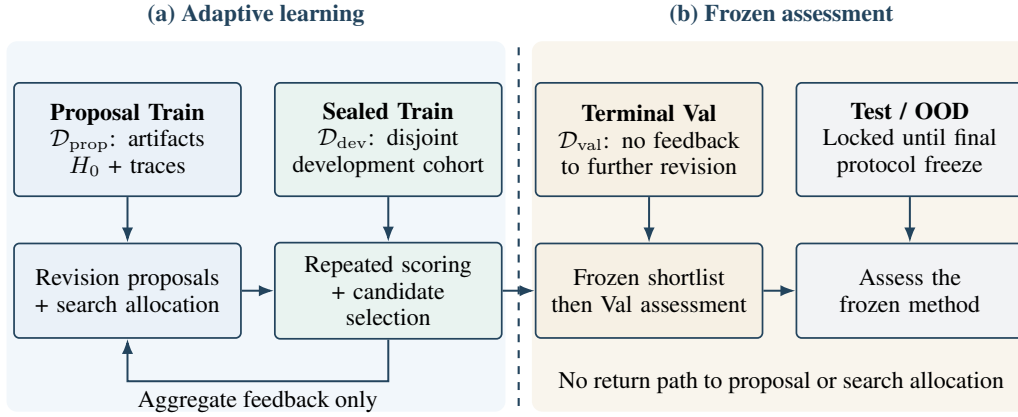
Figure 2 summarizes the evaluation program. An adapter organizes the artifact into task-relevant views, such as rendered slides, document text, or webpage structure. Specialist evaluators apply explicit responsibilities and criteria. They may receive evidence from built-in checks, model-based observations, executable tools, or bounded tool compositions. Routing determines which evidence each specialist receives and may permit bounded follow-up acquisition. Each specialist returns sourced findings and uncertainty alongside its quality assessment. Synthesis then produces the overall judgment.

This representation permits changes that cross an ordinary prompt boundary. For example, a proposed inspection tool is useful only if routing makes its output available to an evaluator that can interpret it. A connected revision may therefore modify the tool, its route, and its consumer together. Specialist responsibilities can also be split, merged, or removed. The search remains bounded by component and edit constraints; it does not assume unrestricted inter-agent communication or recursive self-modification.

4.2 FROM EXECUTION EVIDENCE TO A REVISION

The proposal process examines human-preference disagreements together with correctly ranked protection examples from $\mathcal{D}_{\text{prop}}$. It can inspect the original artifacts and repeated traces, including observations, specialist judgments, and synthesis. Compact case summaries support navigation but do not replace source evidence. The analysis asks where a mismatch arose: whether relevant evidence was absent, a criterion was misapplied, a responsibility was poorly assigned, or the final synthesis made an inappropriate trade-off.

A proposal specifies a mechanism hypothesis, plausible competing explanations, the components to revise, and predicted effects on target and protection cases. The resulting edit is bounded in component scope and text replacement. These constraints limit the size of a revision, not its semantic effect. A connected multi-component edit is measured as a single candidate; an improvement in that candidate does not establish the contribution of every constituent edit.



Original tasks are disjoint across cohorts. Evaluation labels stay outside the pointwise scorer.

Figure 3: Data access during revision and assessment. Panel (a) separates raw proposal evidence from sealed training development measurements, which return aggregate feedback only. Panel (b) assesses a frozen shortlist on Val without feeding its cases or results into further revision; Test/OOD assess the frozen method. Labels used by measurement never enter the pointwise evaluator.

4.3 MEASUREMENT AND SEARCH ALLOCATION

ERA separates three decisions: proposing a change, measuring the resulting program, and allocating further search effort. Local proposal-set evaluations help assess the stated mechanism. They do not serve as a mandatory success filter before development measurement. The latter evaluates frozen candidates on the same sealed $\mathcal{D}_{\text{dev}}$ cohort using Equation 2. Only aggregate development statistics return to the optimizer (Figure 3).

Search maintains a best development program and a working program for the current hypothesis. A new direction starts from the best program; a continuation may retain a completely evaluated intermediate program even when it has not improved the best score. This distinction permits connected revisions over multiple steps without treating every intermediate as a deployment improvement. Promotion requires strictly better development agreement, nondecreasing coverage, complete execution, and a prespecified guard against excessive stratum-level regression. It is a selection rule, not a statistical significance test. Allocation can continue a direction, acquire additional authorized evidence, or start another direction. Algorithm 1 gives the resulting abstraction.

The terminal shortlist contains only complete candidates previously promoted using training development evidence; the default shortlist size is one. If there is no eligible improvement, the seed is retained without querying Val. Terminal assessment does not restart search or adaptively reorder the shortlist. This boundary matters because repeated access to a nominal validation set would otherwise turn it into another development resource.

4.4 EVIDENCE-AWARE SCORE SYNTHESIS

The representation supports both numerical aggregation and a semantic synthesis variant. In the latter, specialists express quality using explicit anchors. Given adjacent anchors with values a_ℓ and $a_{\ell+1}$ and an interval position $\lambda \in [0, 1]$, the score is computed as

$$s = a_\ell + \lambda(a_{\ell+1} - a_\ell), \quad (3)$$

on a scale whose endpoints are 0 and 100. The synthesis module reads sourced findings, reasoning, and specialist scales; it groups overlapping or conflicting factors and locates overall quality on its own anchored scale. It does not assume that raw specialist scores measure the same quantity.

Each finding must be referenced exactly once across synthesis groups. This structural requirement exposes omissions and repeated references, but does not guarantee correct semantic deduplication or factual grounding. Numerical aggregation remains an explicit comparison condition. Changes to

Algorithm 1 ERA: separation of proposal, measurement, and allocation

Require: Seed program θ_0 ; $\mathcal{D}_{\text{prop}}$, sealed $\mathcal{D}_{\text{dev}}$; fixed search constraints

- 1: Measure θ_0 on $\mathcal{D}_{\text{dev}}$; set $\theta_{\text{best}} \leftarrow \theta_0$
- 2: Initialize direction history and the working program
- 3: **while** the prespecified search budget permits continuation **do**
- 4: Choose continuation, additional evidence, or a new direction
- 5: **if** additional evidence is requested **then**
- 6: Acquire authorized proposal evidence; update history; **continue**
- 7: **end if**
- 8: Set parent to the working program (continuation) or θ_{best} (new)
- 9: Form a revision hypothesis from $\mathcal{D}_{\text{prop}}$ and versioned execution evidence
- 10: Construct bounded candidate θ' and verify its execution contract
- 11: Evaluate local predictions and measure θ' on sealed $\mathcal{D}_{\text{dev}}$
- 12: Record the attempt, including incomplete or unfavorable outcomes
- 13: **if** θ' satisfies the development promotion rule **then**
- 14: $\theta_{\text{best}} \leftarrow \theta'$
- 15: **end if**
- 16: Update the working program and direction history for the next allocation
- 17: **end while**
- 18: On normal completion, freeze the eligible shortlist before any Val assessment
- 19: **return** θ_{best} and the complete search record

semantic synthesis require fresh model executions and are charged to the same measurement budget as other program changes.

5 EXPERIMENTAL DESIGN

The experiments are designed to test four questions: whether revision improves held-out human-preference agreement; whether the resulting scores select better artifacts from a fixed pool; whether evaluator feedback improves new artifacts; and which editable components account for any observed effect. The final datasets, sample sizes, model revision, and budgets must be fixed before the confirmatory runs. Appendix C records these outstanding choices.

Domains and supervision. We evaluate the revision method under distinct artifact-evaluation standards, using a domain-specific adapter and evaluator for each standard. Candidate applications include presentations, webpages, and other structured artifacts. Candidate domains and relevant public resources are listed in Appendix B; inclusion is conditional on authorized artifacts and suitable human judgments. Screenshot reconstruction and specification-driven web generation are treated as distinct domains because their evaluation standards differ, despite sharing an output format. Each split is grouped by original input, with near-duplicate sources audited before any optimization. Overall preference is kept distinct from rubric scores and model-generated explanations.

Baselines and controls. The planned comparisons comprise the seed evaluator, a fixed multi-criterion evaluator, prompt-only adaptation, and full program revision. The prompt-adaptation comparison includes an established reflective optimizer such as GEPA, with a prespecified version and edit space. Program-search comparisons expose comparable component interfaces to isolate the effect of the revision procedure. All matched comparisons use the same base model, training supervision, candidate pools, and measurement protocol. Search and deployment costs are reported separately, including tool and synthesis calls. A larger inference budget is not attributed to revision quality.

Preference agreement and selection. The primary endpoint is strict-pair agreement on locked test inputs, reported with coverage, predicted-tie frequency, and source-grouped uncertainty intervals. Paired differences use the same test inputs and repeated-scoring protocol. For best-of- N selection, every evaluator ranks the same prespecified candidate pool for each input. Human assessment of the

Table 1: Planned component comparisons. These are study conditions, not measured findings. Each comparison requires matched supervision, base model, and budgets.

Condition	Change relative to full revision	Question
Prompt-only revision	Freeze tools, routing, and program structure	Does the editable program add value?
Fixed evidence access	Freeze tools and their routing	Does evidence acquisition matter?
Fixed synthesis	Preserve the seed aggregation rule	Does synthesis revision matter?
Numerical synthesis	Replace semantic synthesis with declared numerical fusion	Are the aggregation assumptions consequential?
Restart-only search	Start each proposal from the best program	Does hypothesis continuation help?

Table 2: Primary results template, to be populated separately for each frozen domain and test cohort. A dash means *not measured*. Agreement and coverage are percentages; cost is the mean number of model calls per artifact under the declared repeated-scoring protocol. No ranking or best value is implied.

Evaluator	Agreement	Coverage	Δ vs. seed (95% CI)	Calls
Seed	—	—	—	—
Fixed multi-criterion	—	—	—	—
Prompt-only adaptation	—	—	—	—
ERA	—	—	—	—

selected outputs is the endpoint; existing pair-only labels do not establish best-of- N quality. Neither the pool size nor evaluation cohort is changed in response to a result.

Feedback-guided refinement. For each unseen input, a fixed generator first produces one fresh initial artifact shared by all refinement arms. The seed and evolved evaluators then provide feedback under matched budgets to the same refinement generator. New blinded human comparisons assess the resulting artifacts; labels attached to the original training pairs cannot evaluate these newly generated outputs. Generator settings, refinement rounds, presentation order, and annotation procedures are fixed in advance.

Ablations and reliability. The principal ablations remove component families or alter synthesis while holding the remaining protocol fixed (Table 1). Repeated runs of a fixed evaluator estimate stochastic variability, while multiple independent search seeds assess search variability. Held-out generators, source distributions, or quality regimes provide prespecified OOD tests. These analyses distinguish a stronger proposal mechanism from more favorable selection noise or additional computation.

6 RESULTS

Quantitative evaluation has not been incorporated into this draft. Table 2 reserves the primary reporting structure; all entries denote unmeasured quantities, not zero values. The completed section will report domain-specific estimates before any cross-domain summary, followed by selection, refinement, and component comparisons. Claims of improvement will depend on held-out measurements and uncertainty estimates, not on the best development score encountered during search.

7 DISCUSSION AND LIMITATIONS

Program revision increases the number of possible explanations for an observed change. A connected edit may improve agreement without establishing which component caused the effect; targeted ablations and artifact-level evidence are needed for attribution. Human overall preference also underdetermines the underlying criteria. An evaluator can reproduce an ordering for the wrong reason, and source-linked findings alone do not establish faithful reasoning.

Pointwise scores make selection and refinement possible, but assume that a context-independent quality ordering is useful within the domain. Contextual preferences or incompatible quality dimensions may violate this assumption. Repeated scoring reduces sensitivity to individual calls without eliminating model bias. Adaptive development selection can still overfit, making locked test data necessary.

Finally, tool access, human annotation, and repeated evaluation can be costly. The proposed comparisons must therefore report both agreement and resource use. Cross-domain transfer, improved refinement, and robustness remain open empirical questions. The present scope does not establish benefits for reward-guided parameter training or unrestricted autonomous agent design.

8 CONCLUSION

Learning an evaluator from human preferences requires specifying how evidence is obtained and how quality judgments are combined. We formulate this learning problem over executable evaluation programs and introduce ERA as the revision method. It connects human overall preferences to changes in representation, criteria, specialist responsibilities, evidence acquisition, routing, and synthesis while preserving pointwise deployment. The experimental protocol separates adaptive development from terminal and held-out assessment, and evaluates the scorer both as a judge and as a source of feedback. Whether these revisions yield reliable gains remains to be established by the planned controlled experiments.

AI USE STATEMENT

Generative AI assisted the initial manuscript drafting, literature organization, and preparation of the LaTeX conceptual diagrams. GPT Image 2 generated the conceptual overview illustration, with targeted edits to its information flow. Its artifact miniatures are synthetic illustrations, not observed cases. No empirical observations, human annotations, or performance measurements were generated for this draft. The bibliography was assembled from primary metadata sources and checked for consistency with the cited works. Author review, responsibility for the final content, and the final disclosure of research-stage AI use remain to be completed before submission. This is a draft disclosure, not a declaration that all author verification has already occurred.

ETHICS STATEMENT

The proposed study uses human preferences to develop evaluators, which can encode annotator and model biases rather than a universal definition of quality. Any new annotation study must document consent, compensation, privacy protection, and the applicable institutional review determination before data collection. Artifact licenses and source permissions must be checked independently of technical accessibility. Generated code and evidence tools require restricted execution and controlled access to external resources. No private artifacts, identifiable annotation records, or experimental traces are included in this manuscript skeleton. Study-specific approvals and release decisions are pending.

REPRODUCIBILITY STATEMENT

Sections 3–5 define the program representation, measurement rule, and intended controls. Appendix A specifies completeness and uncertainty reporting; Appendix C lists the configuration items needed for a frozen experiment. The current manuscript contains no results to reproduce. Before submission, the authors must supply a versioned, appropriately anonymized implementation and the permitted data-processing and analysis artifacts, together with completed experiment manifests. Their availability is not asserted here.

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A MEASUREMENT AND ANALYSIS PROTOCOL

Completeness and denominators. For a prespecified strict-preference cohort of N pairs, let $c_i = 1$ when both artifacts have all R valid complete scores. Let $m = \sum_i c_i$ and $k = \sum_{i:c_i=1} z_i$, with z_i from Equation 2. Report completed-pair accuracy k/m when $m > 0$, coverage m/N , and the missing-outcome interval

$$\left[\frac{k}{N}, \frac{k + N - m}{N} \right]. \quad (4)$$

This interval bounds full-cohort agreement under arbitrary missing outcomes; it is not a statistical confidence interval. A missing repeat is not silently replaced by the median of surviving calls. Execution completeness includes selected routes and tools, not just the availability of a final score. A valid observation that an artifact fails a quality check is distinct from a tool execution error.

Randomness and repeated use. Freeze $R \geq 3$, score precision, tie handling, outcome-blind retry policy, and artifact reuse rules before evaluation. Repeated measurements of the same artifact are not independent task observations. Paired uncertainty estimates should resample original input groups, preserving their artifacts and comparisons; the planned interval level is 95%. Report the number of independent search seeds separately from scoring repeats. Repeating calls on the same Val cohort can assess execution stability, but does not create an independent test set.

Selection boundaries. Retain the identities and outcomes of all attempted candidates, including unsuccessful and incomplete attempts, for the authorized internal analysis. Previously measured identical programs are not new independent candidates. Only aggregate sealed-development measurements inform search. The validation shortlist and its order are fixed from training evidence before Val access; all shortlisted outcomes are reported. An interrupted search does not initiate additional terminal assessment. The final test program and any choice among a multi-candidate shortlist must be prespecified without test-driven selection.

B CANDIDATE EVALUATION DOMAINS

Table 3: Candidate domains and their evaluation standards. The table identifies study options, not completed experiments or confirmed dataset inclusion. Existing metrics and rubric labels do not automatically provide overall human pairwise supervision.

Domain	Quality standard	Relevant resources
Slide generation	Source fidelity, narrative, readability, visual organization	PPTAgent; PresentBench
Screenshot-to-web	Faithful reconstruction of a reference rendering	Design2Code
Specification-to-web	Satisfaction of textual and interaction requirements	Corpus and human protocol to be fixed
Summarization	Faithfulness, relevance, coherence, concision	SummEval; human-feedback summarization
Text-to-image	Prompt alignment and human visual preference	Pick-a-Pic; ImageReward; GenEval

For each included domain, document source permissions, original-input grouping, artifact generators, annotation provenance, human agreement, and the availability of task references. Public benchmark examples must not be presented as private newly collected annotations, and benchmark-specific labels must remain outside the deployed scorer’s input. Best-of- N requires a fixed multi-artifact pool and appropriate human assessment beyond any pair-only corpus.

C EXPERIMENT MANIFEST AND COMPLETION CHECKLIST

Table 4: Information required before confirmatory evaluation. Unset fields are intentional placeholders and must not be inferred from development defaults.

Item	Required specification
Data and inclusion	Dataset versions, permissions, domains, input counts, pair counts, exclusions, human-tie policy: <i>TBD</i>
Partitioning	Original-input grouping, deduplication, split sizes, hashes, OOD definition: <i>TBD</i>
Models	Frozen scorer, proposer, tool-model and generator identities, revisions and sampling settings: <i>TBD</i>
Program and search	Seed program, editable components, edit bounds, allocation policy, development guard, shortlist size: <i>TBD</i>
Measurement	$R \geq 3$, precision, completeness, retries, cache identity, search seeds, stopping rule: <i>TBD</i>
Budgets	Search opportunities, model calls, tokens, tool calls, runtime, deployment cost and budget-matching rule: <i>TBD</i>
Downstream studies	Candidate-pool size, shared fresh initial artifacts, refinement rounds, blinded annotation protocol: <i>TBD</i>
Analysis	Primary contrasts, grouped interval computation, multiplicity, missingness analysis and reporting cohort: <i>TBD</i>

The completed paper should replace the primary placeholder table with source-traceable measurements, add paired selection and refinement results, and include representative authorized examples with clear provenance. Any empirical figure must be generated from the frozen analysis data. The two current figures describe the method only.